

Anlage 5

Modulhandbuch des Studiengangs

Electrical Engineering and Information Technology - international Master of Science

des Fachbereichs Elektrotechnik und Informationstechnik
der Hochschule Darmstadt – University of Applied Sciences

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Common Modules

M01 Technical Management

1	Module Name Technical Management
1.1	Module Identifier M01
1.2	Module Type Mandatory
1.3	Course Names Project Management - Lecture Engineering Responsibility - Lecture
1.4	Semester 1 or 2 (winter term)
1.5	Module Responsible and Instructor Prof. Dr. Fromm
1.6	Additional Instructors Prof. Dr. Haid, NN SuK
1.7	Study Program Master / all majors
1.8	Teaching Language English
2	Module Content <u>Content of course "Project Management"</u> This course provides an introduction to professional project management. It covers the areas - introduction into industry process models, e.g. CMMi and SPICE, - project structure, phases, roles and workflow, - relevant methods for requirements engineering, concept development, realization and testing - planning and estimation methods, - risk management, - project tracking metrics, - team building and team management, - change and configuration management, - quality assurance and reviews, - agile methods like SCRUM.

	<u>Content of course "Engineering Responsibility"</u> This course provides an introduction into legal aspects of engineering and discusses the aspect of engineering responsibility. It covers the areas - legal and ethical aspects of engineering responsibility - relevance of penal law, civil law and liability - patent rights - employment law - special liability for safety and security systems - relevant differences in German, European and international laws
3	Learning Outcome / Competencies to understand: - the most relevant laws related to engineering, such as - o Intellectual Property rights (copyrightable and patentable subject matter, infringement of IP rights) - o Contract, labor and data privacy law - o Liability and warranty (including product liability, torts, misdemeanors and crimes, breach of contract) - the differences in German, European and international laws to apply: - industrial engineering and management processes - modern methods of project management - planning and estimation techniques - risk management techniques - configuration and change management techniques - quality assurance methods like reviews - professional team communication techniques - rules of law to product management to transfer: - the project management techniques into new project scenarios - legal awareness into product development and utility scenarios
4	Course Organization and Structure lecture (V) / seminar (S)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching. Project Management: 2 SWS V Engineering Responsibility: 2 SWS V
6	Examination Modalities Examination Prerequisites: none Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes

7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO)
10	Applicability /Utilization This module is applicable for all majors.
11	Literature The following literature material will be provided: - Electronic lecture notes Further literature recommendations will be provided during the lecture.

M02 Team Project

1	Module Name Team Project
1.1	Module Identifier M02
1.2	Module Type Mandatory
1.3	Course Names Team Project - Project
1.4	Semester 1 or 2 (summer term)
1.5	Module Responsible and Instructor Prof. Dr. Fromm
1.6	Additional Instructors All instructors of the master's program
1.7	Study Program Master / all majors
1.8	Teaching Language English
2	Module Content Content of course "Team Project" In this course, the students execute a practical project using the methods presented in the module "Technical Management". It covers the areas: - practical development of a technical system (project work) - project management and work package agreement, - requirements engineering, system design, - implementation and testing, - team building and team communication, - and documentation and presentation of the results.
3	Learning Outcome / Competencies to understand: - project roles, phases and workflows to apply: - workpackage definition and assignment - modern methods of project management and engineering

	- professional team communication techniques - project presentation to transfer: - the project management and engineering techniques into new project scenarios
4	Course Organization and Structure Project
5	Credits and Workload 5 CP / 150 hours in total, project work including meetings with the instructors.
6	Examination Modalities The examination consists of the project work and the project presentation. As a consequence, a withdrawal from the project is only possible once within the first 2 weeks of the project. A later withdrawal will be marked as "not passed". For details, please check § 13 para. 5 BBPO. Examination Prerequisites: Successful fulfillment of prerequisites are measured by: - Attending Project - Project Progress Examination Type: - Project presentation and Report - Milestone review Examination Duration: Project presentation 30 minutes
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO).
10	Applicability /Utilization This module is applicable for all majors.
11	Literature Literature recommendations will be provided during the project.

M03 Internship

1	Module Name Internship
1.1	Module Identifier M03
1.2	Module Type Mandatory / 4 semester course
1.3	Course Names German Class 1 and 2 – Lecture Preliminary Seminar – Seminar Internship
1.4	Semester - German classes are offered during winter semester (extensive phase). Additional intensive courses will be offered in the month September and February/March (intensive phase). - The internship generally takes place in the third semester of the 4 semester course. The internship is offered in both winter and summer semester.
1.5	Module Responsible and Instructor Prof. Dr. Krauß, Head of the „Sprachenzentrum“ (FB GW)
1.6	Additional Instructors All instructors of the master's program can act as academic supervisor for the internship part. instructors of the „Sprachenzentrum“
1.7	Study Program Master / all majors
1.8	Teaching Language English
2	Module Content Content of German Class: - German Class 1: A1 level or higher - German Class 2: higher than German Class 1 level, at least A2 level Content of Preliminary Seminar: - Preparative items (such as regulations and application matters) are presented. Content of Internship: The student has to solve an engineering task in the area of electrical engineering and information technology under the guidance of an industrial supervisor and an academic supervisor. This internship work can involve one of the following areas:

	- Research and development work - Project planning and design - Manufacturing, preparation of work - Assembly - Test bed, quality control
3	Learning Outcome / Competencies The objectives of the internship are as follows: - Create a linkage between the studies and the world of work - Orientation in the profession strived for - Get to know technical and organizational contexts - Involvement in the process of work - Practical training in the engineering trade in one or several projects.
4	Course Organization and Structure Lecture, seminar, practical work by fulfilling tasks of engineering work and documentation
5	Credits and Workload 30 CP / 900 hours in total - German class 1 and 2: 6 SWS in total, 84 hours classroom teaching in total. Additional intensive preparation courses will be offered during semester breaks. - Internship: 26 weeks full-time work in a company
6	Examination Modalities German Classes Examination Prerequisites (German Class 2): German Class 1 has been successfully passed. Examination type: A combined written and oral examination will be offered in each class. Internship Examination Prerequisites: The Preliminary Seminar must have been attended and at least German Class 1 (level A1 or higher) must have been successfully passed before the start of the internship. Examination Type: Students must prepare a technical report on their internship work with a range of 30 to 40 pages. The report shall be submitted to the academic supervisor at the latest 2 weeks after the work has ended (and otherwise at the latest before the master thesis starts). The report will be assessed and evaluated by the academic supervisor. The successful completion of the internship part is acknowledged by the academic supervisor, provided that the prerequisites defined in § 10 para. 4 are fulfilled.
7	Necessary Prerequisites The necessary prerequisites are defined in § 10 para. 2 BBPO
8	Recommended Prerequisites A total of 45 CP is recommend to be completed before the start of the internship work.

9	Duration and Frequency of Course The internship module consists of the internship part (practical work) itself and preparatory lectures. The internship part lasts 26 weeks. It may be undertaken any time.
10	Applicability /Utilization The module provides the prerequisites for the Master thesis. It is applicable for all majors.
11	Literature ---

M04 Master Module

1	Module Name Master Module
1.1	Module Identifier M04
1.2	Module Type Mandatory
1.3	Course Names Thesis Colloquium
1.4	Semester 3 semester course: 3 (winter term) 4 semester course: 4 (summer term)
1.5	Module Responsible and Instructor Head of the examination board
1.6	Additional Instructors All instructors of the master's program
1.7	Study Program Master / all majors
1.8	Teaching Language N/A
2	Module Content - Practically and/or theoretically oriented scientific work in the area of the chosen major - Written thesis - Colloquium
3	Learning Outcome / Competencies Students should demonstrate the following qualifications within the area of the defined topic: - Capability of independent work - Systematic analysis and solutions using engineering and scientific methods - Professional competence in scientific documentation and presentation of results
4	Course Organization and Structure final thesis (FT)

5	Credits and Workload 30 CP / 900 hours in total.
6	Examination Modalities Examination type: Master thesis under § 12 para. 4 and 5 BBPO Colloquium under § 12 para. 6 and 7 BBPO The assessment ratio of thesis and colloquium are defined in § 12 para. 8 BBPO.
7	Necessary Prerequisites The necessary prerequisites are defined in § 12 para. 3 BBPO.
8	Recommended Prerequisites All mandatory modules of study program
9	Duration and Frequency of Course The master thesis must not exceed 6 months. With the approval of the examination board, the master thesis may be undertaken at any time.
10	Applicability /Utilization This module is applicable for all majors.
11	Literature Will be recommended by supervisors.

MWP01 Elective 1**MWP02 Elective 2****MWP03 Elective 3**

1	Module Name Elective 1 Elective 2 Elective 3
1.1	Module Identifier Major „Automation“: MAWP01, MAWP02, MAWP03 Major „Communications“: MCWP01, MCWP02, MCWP03 Major „Embedded and Microelectronics“: MMWP01, MMWP02, MMWP03 Major „Power Engineering“: MPWP01, MPWP02, MPWP03
1.2	Module Type Elective
1.3	Course Names Students have to choose at least 15 CP from the catalogue of their own major. The catalogues contain major specific subjects as well as subjects from other majors which are suitable for the certain major Major „Automation“: MAwp Major „Communications“: MCwp Major „Embedded and Microelectronics“: MMwp Major „Power Engineering“: MPwp For general rules regarding electives please check § 9 BBPO. To get an overview of the catalogues contents see appendix 2 BBPO. Descriptions of the elective subjects are included in this handbook (appendix 5 BBPO).
1.4	Semester summer term and winter term (see appendix 1 BBPO Study program)
1.5	Module Responsible and Instructor see descriptions of the elective subjects
1.6	Additional Instructor see descriptions of the elective subjects
1.7	Study Program Master / all majors
1.8	Teaching Language see descriptions of the elective subjects

2	Module Content see descriptions of the elective subjects
3	Learning Outcome / Competencies see descriptions of the elective subjects
4	Course Organization and Structure check descriptions of the elective subjects
5	Credits and Workload each module (Elective 1 to 3): 5 CP / 150 hours in total see descriptions of the elective subjects for the number of hours classroom teaching (SWS)
6	Examination Modalities see descriptions of the elective subjects
7	Necessary Prerequisites None
8	Recommended Prerequisites see descriptions of the elective subjects
9	Duration and Frequency of Course Each elective subject will take one semester and may be offered once a year, but the department is not liable to offer all of them. A list of elective subjects will be published at the beginning of each semester.
10	Applicability /Utilization see descriptions of the elective subjects
11	Literature see descriptions of the elective subjects

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Common Elective Modules (general electives)

MGwp01 Research Project

1	Module Name Research Project
1.1	Module Identifier MGwp01
1.2	Module Type Elective
1.3	Course Names Research Project
1.4	Semester summer term or winter term
1.5	Module Responsible and Instructor Prof. Peter Fromm
1.6	Additional Instructor All instructors of the master's program
1.7	Study Program Master / all majors
1.8	Teaching Language English
2	Module Content In this course, the students execute a research focused project at the university. It covers the areas: - Introduction to research methodologies - Work on state of the art research questions - Development of a research strategy - Execution of the research work - Presentation of the results
3	Learning Outcome / Competencies to understand: - state of the art research topics to apply: - research methodology - professional team communication techniques - project presentation to transfer: - research techniques into new project scenarios

4	Course Organization and Structure Project
5	Credits and Workload 2,5 CP / 75 hours in total
6	Examination Modalities The examination consists of the project work and the project presentation. As a consequence, a withdrawal from the project is only possible once within the first 2 weeks of the project. A later withdrawal will be marked as "not passed". For details, please check § 13 para. 5 BBPO. Examination Prerequisites: Successful fulfillment of prerequisites are measured by: - Attending Project - Project Progress Examination Type: - Project presentation and Report - Milestone review Examination Duration: Project presentation 30 minutes
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course The elective will be offered based on availability of topics.
10	Applicability /Utilization All majors
11	Literature Project specific

MGwp02 Selected Research Topics

1	Module Name Selected Research Topics
1.1	Module Identifier MGwp02
1.2	Module Type Elective
1.3	Course Names Selected Research Topics
1.4	Semester summer term or winter term
1.5	Module Responsible and Instructor NN
1.6	Additional Instructor All instructors of the master's program
1.7	Study Program Master / all majors
1.8	Teaching Language English
2	Module Content In this course, the students will be introduced to selected state of the art research topics in the following areas - Automation, robotics, factory automation and (II)IoT - Communication systems and technologies, signal processing - Embedded systems, hardware and software engineering, SOC and FPGA development - Power generation and distribution, advanced control systems, e-mobility - Other topics upon availability
3	Learning Outcome / Competencies to understand: - Research methodologies to apply: - Working with literature and other information sources - Scientific writing skills - Scientific presentation skills and discussions

	to transfer: - The gained knowledge to own research work, e.g. master thesis
4	Course Organization and Structure lecture (V) 2 SWS V
5	Credits and Workload 2,5 CP / 75 hours in total
6	Examination Modalities Examination Prerequisites: - none Examination Type: - Scientific report and presentation followed by an oral colloquium Examination Duration: 45 min
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03) The elective will be offered based on availability of topics.
10	Applicability /Utilization All majors
11	Literature Topic specific

MGwp03 Ethics, Engineering, and Ecology

1	Module Name Engineering Ethics
1.1	Module Identifier MGwp03
1.2	Module Type Elective
1.3	Course Names Ethics, Engineering, and Ecology
1.4	Semester summer term or winter term
1.5	Module Responsible and Instructor Prof. Dr. Bernd Steffensen (Dep. Social Sciences)
1.6	Additional Instructor ---
1.7	Study Program Master / all majors
1.8	Teaching Language English
2	Module Content This course provides an introduction to the topic of ethics in engineering with a special focus on challenges provided by ecological requirements. Future developments and the well-being of the world society will depend on technical solutions. The development and the application of technical artifacts has numerous impacts beyond pure functioning. Therefore the course will cover the areas: - Why engineering ethics? - Engineering and Responsibility - A Matter of Moral - Individual Moral and the Organizational Context - Honesty, Integrity, and Reliability - Safety, Risk, and Liability - Solving Ethical Problems - Engineering and Environment - Ethics, Competing Values, and Stakeholders - Professionalism

3	Learning Outcome / Competencies to understand: - the importance of ethics in engineering - the basic philosophical concepts of an applied ethic - the competing, nevertheless, appropriate values of various stakeholder in society - the future impact of Sustainable Development Goals (SDGs) and environmental concerns for engineering in general and firms in particular. to apply: - theories and methods to identify ethical issues in real world situations. - theories and methods to develop an individual position and to argue accordingly to transfer: - ethical reasoning to various contexts in the workplace and the fields of individual engagement in the civil society.
4	Course Organization and Structure seminar (S)
5	Credits and Workload 2,5 CP / 75 hours, including 28 hours (20 hours on a weekly base, 8 hours as a block course on a single day) 2 SWS seminar
6	Examination Modalities Examination Prerequisites: - none Examination Type: - Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: - 90min
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03)
10	Applicability /Utilization All majors

11	Literature The following literature material will be provided: - Harris, Ch.E., Jr./Pritchard, M.S./James, R.W./Englehardt, E.E./Rabins, M.J. (2019): Engineering Ethics, Concepts and Cases, Cengage, Boston On the learning platform operated by the Hochschule Darmstadt, an updated course is created for each semester, in which literature references are given and the necessary material is provided as far as possible.
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MGwp04 Legal English for Engineers

1	Module Name Legal English for Engineers
1.1	Module Identifier MGwp04
1.2	Module Type Elective
1.3	Course Names Legal English for Engineers
1.4	Semester Summer or winter term
1.5	Module Responsible and Instructor Dr. Tamara Onnis (Language Center Coordinator of Legal English)
1.6	Additional Instructors ---
1.7	Study Program Master / all majors
1.8	Teaching Language English
2	Module Content This course provides an introduction to the topic Legal English for Engineers. It includes: Legal aspects of doing business • Right and responsibilities of manufacturers under contracts • General and specific Liabilities issues in Industry agreements • Patents, design rights copyrights (Including international licensing law) Contracts • Types of legal Agreements • Reading and understanding service agreements and manufacturing contracts • Legal jargons for technicians Safety and security • Management processes, requirements and regulations

	• Current and future legal framework (of eg. autonomous driving and vehicle design) in European countries and North America • Legal responsibilities 1. Government, 2. Manufacturer 3. Consumer
3	Learning Outcome / Competencies This course will help students develop both a theoretical and practical understanding of legal English in an Engineering context. to know: • How to read, analyze and conform to legal framework and requirements to understand: • How to understand basic and technical legal agreements as they relate to the field of Engineering to apply: • How to utilize legal jargons and terminology in spoken and written form in practice to transfer: • How to be able to transfer legal knowledge in daily tasks and services.
4	Course Organization and Structure The course is structured as a seminar either in block form or on a weekly basis. It is based on lectures, projects, exercises and group work.
5	Credits and Workload 2.5 CP, including 34 hours classroom teaching and 41 hours independent learning 2SWS V
6	Examination Modalities Possible changes to the examination modalities may be communicated upon start of the module. Examination Prerequisites: In order to write the end-of term exam, it is required to successfully attend minimum 75%/ of all class, successfully participate in class exercises and carry out a presentation. Successful fulfillment of prerequisites are measured by: - presentation - exercises - attending classes - group work - projects Examination Type: • Written exam 50% • presentation 30% • homework/assignments/participation 20%

	Examination Duration: • 90 Minutes
7	Necessary Prerequisites None
8	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03)
10	Applicability /Utilization All majors
11	Literature Learning material will be provided during the course.

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Major Automation - mandatory

MA01 Safety in Industrial Automation

1	Module Name Safety in Industrial Automation
1.1	Module Identifier MA01
1.2	Module Type Mandatory
1.3	Course Names Safety in Industrial Automation – Lecture Safety in Industrial Automation - Lab
1.4	Semester 1 or 2 (winter term)
1.5	Module Responsible and Instructor Prof. Dr.. Simons
1.6	Additional Instructors ---
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content Content of course “Safety in Industrial Automation - Lecture”: Participants will be exposed to and gain working experience to design, implement, verify and validate safe systems in industrial automation. The course will cover - Introduction to safety systems in industrial automation - Basic terminology and standards concerning safety in industrial automation - Design of safe control systems - Measures to achieve safe processes and safe products incl. overview of safety devices - Development of safety related software using safety PLCs: configuration & programming of safety PLCs - Verification & validation of safety measures in accordance to a standard, e.g. EN ISO 13849 Content of course “Safety in Industrial Automation - Lab”: Practical design and programming of safety programs using safety PLCs and practical verification of safety in accordance to a standard are part of this course: - Hardware configuration of safety PLCs

	- Design, implementation, testing and debugging of a safety program using a safety PLC - Calculating the safety level required, e.g. performance level required in accordance to EN ISO 13849 - Verification of safety systems e.g. by calculating the performance level achieved in accordance to EN ISO 13849
3	Learning Outcome / Competencies to understand: - the basics of safety in industrial automation - the basic standards and the terminology for safe systems in industrial automation - the different concepts to achieve safe systems - the structure of safety systems - the verification and validation process for safe systems to apply the gained knowledge: - to design safe control systems, i.e. to design the safety concept, to select meaningful safety devices, to implement the hardware of the system including where to place the safety devices and the electrical connection - to develop safety related software using safety PLCs including the hardware configuration, the programming, the testing and the debugging - to verify and validate safe systems in industrial automation in accordance to a standard, e.g. EN ISO 13849 to transfer: - the acquired knowledge to create, verify and validate safe systems in industrial automation.
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V / 0,5 (1) SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: - Attending Lab - Lab Progress Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes.
7	Necessary Prerequisites Participation in the preliminary course if no programming knowledge of Siemens S7 PLCs is available.
8	Recommended Prerequisites None

9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO).
10	Applicability /Utilization This module is applicable for the major Automation.
11	Literature The following literature material will be provided: - Electronic lecture notes (excerpt of slides) - Workbook for the lab Further literature recommendations will be provided during the lecture.

MA02 Adaptive Control, Modeling and Identification

1	Module Name Adaptive Control, Modeling, and Identification
1.1	Module Identifier MA02
1.2	Module Type Mandatory
1.3	Course Names Adaptive Control, Modeling, and Identification – Lecture Adaptive Control, Modeling, and Identification – Lab
1.4	Semester 1 or 2 (winter term)
1.5	Module Responsible and Instructor Prof. Dr. Kleinmann
1.6	Additional Instructors Prof. Dr. Weigl-Seitz
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content The course covers the areas: - Introduction to and classification of Adaptive Control Systems - Adaptation of Optimal Controllers and Controller Design by Pole Placement - Dynamic Behavior of Adaptive Control Loops and Configuration Issues - Modeling of Linear and Non-linear Time-invariant and -variant Dynamic Systems - Algorithms and Filters for Online Process Identification - Neural Networks as Memory Blocks for Controller and Process Model in Learning Control Loops - Computer based applications using Matlab/Simulink
3	Learning Outcome / Competencies to understand: - the basic applications, concepts, components and challenges of adaptive control loops - the basic principles for modeling and identification of complex dynamic systems

	to apply the knowledge: - to classify a control application and select an appropriate approach for adaptive control - to know the advantages and disadvantages of modeling and identification algorithms and structures - to implement and simulate the components of an adaptive control loop to transfer: - the design process of adaptive control systems to problems from various domains in automation
4	Course Organization and Structure Lecture (V) / Laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V and 0,5 (1) SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: • Attending Lab • Lab Progress Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes.
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO).
10	Applicability /Utilization This module is applicable for the major Automation and for the major embedded systems and microelectronics.

11	Literature The following literature material will be provided: - Electronic Script (excerpt of slides) - Workbook for the lab Further literature recommendations will be provided during the lecture.
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MA03 Computer Vision

1	Module Name Computer Vision
1.1	Module Identifier MA03
1.2	Module Type Mandatory
1.3	Course Names Computer Vision Computer Vision Lab
1.4	Semester 1 or 2 (winter term)
1.5	Module Responsible and Instructor Prof. Dr. Naser
1.6	Additional Instructors Prof. Dr. Neubecker
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content - Image Sensors - Image formation and digital images - 3D Sensors and point clouds - Image enhancement - Object recognition techniques - Pattern classification - Camera calibration - Stereo vision techniques and algorithms - Case studies of selected imaging solutions for Automation, Robotics and Industrial Image Processing

	Learning Outcome / Competencies to understand: - the mathematical and theoretical foundations of image processing and computer vision - the basic components and working principles of 2D- and 3D-Machine Vision Systems - the difference between image and point cloud based approaches to vision problems and their areas of application. - the uses and limitations of computer vision through practical case studies to apply the knowledge: - to know the advantages and disadvantages of different imaging sensors - to select appropriate hardware components for a given imaging scenario - to identify a suitable chain of algorithms for a given imaging problems to transfer: - the knowledge acquired in the lectures to new vision problems in Robotics, Automation and Production.
	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V and 0,5 (1) SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: • Attending Lab • Lab Progress Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes.
7	Necessary Prerequisites None
8	Recommended Prerequisites Linear Algebra, Matlab
9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO).

10	Applicability /Utilization This module is applicable for the major Automation and for the major Embedded Systems and Microelectronics.
11	Literature - Hartley, Zissermann: Multiple View Geometry - Steeger Ulrich Widemann: Machine Vision Algorithms and Applications - Burger, Burge: Digital Image Processing Further literature recommendations will be provided during the lecture.

MA04 Advanced Programming Techniques

1	Module Name Advanced Programming Techniques
1.1	Module Identifier MM01 / MA04
1.2	Module Type Mandatory
1.3	Course Names Advanced programming Techniques - Lecture Advanced programming Techniques - Lab
1.4	Semester 1 or 2 (winter term)
1.5	Module Responsible and Instructor Prof. Dr. Lipp
1.6	Additional Instructors Prof. Dr. Fromm, Prof. Dr. Bürgy
1.7	Study Program Master / Major Embedded Systems and Microelectronics Master / Major Automation
1.8	Teaching Language English
2	Module Content Content of course “Advanced Programming Techniques” Review of fundamental concepts of a widely used object oriented programming language. The course will cover - introduction to the UML - OOA and OOD techniques - class design and class relations in C++, - C++ operator overloading, - advanced data structures, design patterns and algorithms - systematic test techniques Design aspects like modularity and software re-use will be discussed. Developing software designs using the UML and CASE tools as well as extensive hands-on programming assignments in C/C++ are an integral part of the course.

3	Learning Outcome / Competencies to understand: - the fundamentals of professional software design to apply: - the C++ programming language - the UML diagrams to transfer: - the design patterns and concepts to more complex architectures
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 2 SWS V and 2 SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: - attending Lab - code walkthrough during the lab 20% of the module grades are obtained by the laboratory during the term. Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO).
10	Applicability /Utilization This module is applicable for the major Automation and for the major Embedded Systems and Microelectronics.

11	Literature The following literature material will be provided: - Electronic lecture notes - Workbook for the lab Further literature recommendations will be provided during the lecture.
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MA05 Industry 4.0/IIoT and the Digital Factory

1	Module Name Industry 4.0/IIoT and the Digital Factory
1.1	Module Identifier MA05
1.2	Module Type Mandatory
1.3	Course Names Industry 4.0/IIoT and the Digital Factory – Lecture Industry 4.0/IIoT and the Digital Factory - Lab
1.4	Semester 1 or 2 (summer term)
1.5	Module Responsible and Instructor Prof. Dr. Simons
1.6	Additional Instructors ---
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content Content of course “Industry 4.0/IIoT and the Digital Factory”: Participants will be exposed to and gain working experience to design, implement, verify and validate safe systems in industrial automation. The course will cover - Goals and Key technologies for modern production - Basics of Industrie 4.0, IoT and Digital transformation, Structures of modern production - Product identification systems - Data acquisition and exchange, e.g. OPC UA - Product lifecycle management and product data management (PLM / PDM) - Simulation systems (HIL/SIL, virtual commissioning, material flow and energy consumption simulation) - Remote control - Security and safety in industrial automation - Manufacturing execution systems (MES) - Enterprise resource planning systems (ERP) - Assistance systems, e.g. using mixed reality

	- Cloud computing including e.g. IoT hubs, cloud services
3	Learning Outcome / Competencies to understand: - the goals and key technologies of the digital transformation - the structure of modern production - the possibilities, the advantages and the challenges of digitalized production - the basics of security for digitalized companies - the basics of MES, ERP, PLM/PDM, Assistance systems and cloud computing to apply: - implementing appropriate product identification systems - using simulation systems for virtual commissioning - developing ideas for new business processes of digitalized production companies to transfer: Students shall be able to play a significant role in the digital transformation of companies, by being able - to give impulses and develop scenarios for the digitalization of production companies - to choose appropriate information technologies, to judge about the advantages and challenges of these technologies and - to implement the communication between the information technology and the control system.
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V and 0,5 (1) SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: • Attending Lab • Lab Progress Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes.
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO).

10	Applicability /Utilization This module is applicable for the major Automation.
11	Literature The following literature material will be provided: - Electronic Script (excerpt of slides) - Workbook for the lab Further literature recommendations will be provided during the lecture.

MA06 Industrial Robotics

1	Module Name Industrial Robotics
1.1	Module Identifier MA06
1.2	Module Type Mandatory
1.3	Course Names Industrial Robotics – Lecture Industrial Robotics – Lab
1.4	Semester 1 or 2 (summer term)
1.5	Module Responsible and Instructor Dr. Koch
1.6	Additional Instructors
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content The course covers the areas: - Overview on Applications, Systems and Technologies of Industrial Robotic Systems - Kinematic and Kinetic Models for Industrial Robots - Path Planning and Control Algorithms for Industrial Robots - Integration of Sensors and Multimodal Servoing - Force Control and Human-Robot Collaboration - Technological Aspects (e.g., Accuracy, Safety issues, Energy consumption)
3	Learning Outcome / Competencies to understand: - the basic applications, systems, and technologies of industrial robotic systems - the basic principles for the design of a robotic system with regard to a specific task

	to apply the knowledge: - to classify robotic applications and describe the necessary system architecture - to know and simulate the fundamental models and algorithms for industrial robots to transfer: - the control of 6-axis robots to new kinematics and cells with multiple arms
4	Course Organization and Structure Lecture (V) / Laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V and 0,5 (1) SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: • Attending Lab • Lab Progress Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes.
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO).
10	Applicability /Utilization This module is applicable for the major Automation and for the major Embedded Systems and Microelectronics.
11	Literature The following literature material will be provided: - Electronic lecture notes (excerpt of slides) - Workbook for the lab Further literature recommendations will be provided during the lecture.

MA07 State Space Control Design

1	Module Name State-Space Control Design
1.1	Module Identifier MA07
1.2	Module Type Mandatory
1.3	Course Names State-Space Control Design – Lecture State-Space Control Design - Lab
1.4	Semester 1 or 2 (summer term)
1.5	Module Responsible and Instructor Prof. Dr. Weigl-Seitz
1.6	Additional Instructors
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content Content of Course „State-Space Control Design – Lecture“: • Modelling of dynamic systems using state variables • State space representation of dynamic systems • Correlation between transfer functions and state space representation • Structural properties (stability, controllability, observability) • Canonical Forms • State space transformations • Solution of the time-invariant state-space equations • Design of state feedback controllers • Design of state observers • State feedback by optimal control • Computer based applications using Matlab/Simulink Content of Course „State-Space Control Design – Lab“: • Exercises on modelling and designing state-space control systems

	• Practical programming assignments for state-space applications in Matlab/Simulink
3	Learning Outcome / Competencies to understand: • the state-space concept to apply: • solve time-invariant state-space equations • design state-feedback systems by optimal control to transfer : • describe systems using state-space representations • analyze systems using state-space techniques • design state feedback controllers and state observers • programming of state-space applications in Matlab/Simulink
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V / 0,5 (1) SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: - Attending Lab - Exercises and Lab Progress - Documentation / Lab Report Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course This module takes one semester and is offered once a year (see appendix 1 BBPO).

10	Applicability /Utilization This module is applicable for the major Automation
11	Literature The following literature material will be provided: - Electronic lecture notes and exercises Further literature recommendations will be provided during the lecture.

Module Handbook

Electrical Engineering and Information Technology - international

Master of Science

Major Automation - electives

MAwp01 Model-based Real-time Simulation of Mechatronic Systems

1	Module Name Model-based Real-time Simulation of Mechatronic Systems
1.1	Module Identifier MAwp01
1.2	Module Type Elective
1.3	Course Names Model-based Real-time Simulation of Mechatronic Systems – lecture Model-based Real-time Simulation of Mechatronic Systems – lab
1.4	Semester 1 or 2
1.5	Module Responsible and Instructor N.N.
1.6	Additional Instructors ---
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content Model-based Real-time Simulation of Mechatronic Systems – lecture This course provides the concepts of model-based real-time simulation and system design. The course covers the areas: - Modelling and classification of mechatronic systems - Application areas, requirements - Real-time simulation and rapid prototyping methods - Hardware-in-the-loop, software-in-the-loop and processor-in-the-loop - Experimental validation and testing methods - Summary, Conclusion and future prospects Model-based Real-time Simulation of Mechatronic Systems – lab This lab provides projects to design model-based real-time simulation and system design. The lab covers the areas: - Introduction MATLAB/SIMULINK

	- Software and function development process - Real-time simulation and rapid prototyping applications - Automatic code generation - Experimental validation and testing methods
3	Learning Outcome / Competencies to understand: - the structure of mechatronic systems - model-based development procedure of mechatronic systems - the improvement of the system's documentation and maintainability to apply: - model mechatronic systems - improve the design and implementation process of mechatronic systems to transfer: - exemplary model-based simulation and testing of mechatronic systems
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V/ 1 SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfilment of prerequisites are measured by: • Attending Lab • Lab Progress Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03)

10	Applicability /Utilization This module is applicable for the major Automation. See appendix 2 BBPO (Compulsory options catalogues) for its suitability for other majors.
11	Literature The following literature material will be provided: - Electronic lecture notes - Workbook for the lab Further literature recommendations will be provided during the lecture.

MAwp02 High Level Language Frameworks

1	Module Name High Level Language Frameworks
1.1	Module Identifier MAwp02
1.2	Module Type Elective
1.3	Course Names --
1.4	Semester 1 or 2
1.5	Module Responsible and Instructor Prof. Dr. Lipp
1.6	Additional Instructors
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content Participants will be introduced to the development of graphical applications using Android and JAVA. The course will cover - JAVA language basics - Threads and synchronization - Framework tools, f.ex. from Android - Graphical user interfaces Practical programming assignments in JAVA and Android will be part of the course.
3	Learning Outcome / Competencies to understand: - the JAVA language - threads - GUI Design - events

	to apply: - high level language, threading and events - interface definitions of a corresponding framework by implementing a small applications using Android.
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V/ 1 SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by lab exercises. Examination type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes
7	Necessary Prerequisites Basic programming skills.
8	Recommended Prerequisites Object oriented programming.
9	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03)
10	Applicability /Utilization This module is applicable for the major Automation. See appendix 2 BBPO (Compulsory options catalogues) for its suitability for other majors.
11	Literature The following literature material will be provided: - Electronic lecture notes - Workbook for the lab Further literature recommendations will be provided during the lecture.

MAwp03 Human Machine Interfaces (HMI)

1	Module Name Human Machine Interfaces (HMI)
1.1	Module Identifier MAwp03
1.2	Module Type Elective
1.3	Course Names Human Machine Interfaces - lecture Human Machine Interfaces - laboratory
1.4	Semester 1 or 2
1.5	Module Responsible and Instructor Prof. Dr. Bürgy
1.6	Additional Instructors Prof. Dr.. Wirth
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content • Human senses • Human perception • Interaction channels between humans and machines • General design aspects • Modeling of user interaction (UML-based architecture design) • Usability / user experience • Machine interfaces (widgets, IO, WIMP and post-WIMP interfaces) • Testing (software, user and field tests)
3	Learning Outcome / Competencies to know: • relevant definitions • design guidelines

	to understand: - mechanisms of human senses - capabilities and restrictions of human perception - foundation of user experience to apply: - usability aspects for designing user interaction with machines - selecting the right interface mechanisms and SW/HW interfaces to transfer: - modeling techniques, especially UML-based interface definition and documentation - choosing the right interface templates for human machine interaction - adapting interaction principles to multi-modal human machine interaction
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 2,5 CP / 75 hours in total, including 28 hours classroom teaching and lab. 2 SWS V
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: - Laboratory Workbook - Attending Lab Examination Type: continuous project covering the complete content of the module with a final presentation at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 15 minutes presentation.
7	Necessary Prerequisites None
8	Recommended Prerequisites Advanced Programming Techniques (MA04)
9	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03)
10	Applicability / Utilization All (software) design tasks; software and hardware architectures; mechatronics, WING. See appendix 2 BBPO (Compulsory options catalogues) for its suitability for other majors.

11	Literature • Dix, F.; Abowd, B.: Human-Computer Interaction, 3rd Edition, Pearson Education Ltd., 2004978-0130461094 • Norman, D.: The Design of Everyday Things, revised and expanded edition, Basic Books, 2013, ISBN: 978-0-465-05065-9 • Scott MacKenzie: Human-Computer Interaction: An Empirical Research Perspective, 2013, Morgan Kaufmann; ISBN: 978-0124058651
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MAwp04 Autonomous Mobile Robots

1	Module Name Autonomous Mobile Robots
1.1	Module Identifier MAwp04
1.2	Module Type Elective
1.3	Course Names Autonomous Mobile Robots – Lecture Autonomous Mobile Robots - Lab
1.4	Semester 1 or 2
1.5	Module Responsible and Instructor Dr. Lübbers
1.6	Additional Instructors
1.7	Study Program Master / Major Automation
1.8	Teaching Language English
2	Module Content Introduction to - Application examples - Locomotion - Kinematics of mobile robots - Perception for mobile robots - Characteristics of mobile robots in structured and unstructured environments - Mobile robot localization methods, algorithms - Planning and navigation, incl. maps, methods for autonomous map generation and obstacle avoidance - Navigation architectures of autonomous mobile robots - Showcase demonstration and validation of methods using laboratory systems

3	Learning Outcome / Competencies to understand: - the basic applications, concepts, components and challenges of autonomous mobile robots - the basic locomotion principles for legged and wheeled robots to apply the knowledge: - to choose an appropriate locomotion concept for a new application - to develop the kinematic model for different mobile robots with wheels - to know the advantages and disadvantages of different sensors for mobile robots and to choose the appropriate sensors for an application - to implement the odometry method for localization to a wheeled mobile robot - to choose appropriate planning and navigation methods and a an appropriate navigation architecture to transfer: - the design patterns and concepts to other autonomous mobile robots for other applications.
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 5 CP / 150 hours in total, including 56 hours classroom teaching and lab. 3 SWS V/ 1 SWS L
6	Examination Modalities Examination Prerequisites: In order to participate in the module exam, it is required to successfully finish the lab part of the module. Successful fulfillment of prerequisites are measured by: - Attending Lab - Lab Progress Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes.
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03)
10	Applicability /Utilization This module is applicable for the major automation. See appendix 2 BBPO (Compulsory options catalogues) for its suitability for other majors.

11 Literature

The following literature material will be provided:

- Electronic lecture notes (excerpt of slides)
- Workbook for the lab

Further literature recommendations will be provided during the lecture.

MAwp05 Advanced Graphical Programming of Control Systems

1	Module Name Advanced Graphical Programming of Control Systems
1.1	Module Identifier MAwp05
1.2	Module Type Elective
1.3	Course Names Advanced Graphical Programming of Control Systems - lecture
1.4	Semester 1 or 2
1.5	Module Responsible and Instructor Prof. Dr. Haid
1.6	Additional Instructors -
1.7	Study Program Master / Major Automation, Embedded and Power
1.8	Teaching Language English
2	Module Content • Modeling of user interaction • LabVIEW Queued Message Handler • One-to-many communication . • Managing Software Engineering in Graphical Languages • Advanced Architectures for Graphical Languages in LabVIEW • Object-Oriented Design and Graphical Programming • Performance of Graphical Languages
3	Learning Outcome / Competencies to know: • Realization of applications with the help of graphical programming languages • Different architectural approaches to understand: • Special features of graphical programming • Use of graphical libraries • Validation of graphical software applications

	to apply: • Develop a customized user interface prototype for initial usability testing. • Handle errors locally or globally depending on the severity of the error. • Develop, integrate, and test scalable, readable, and maintainable code modules. to transfer: • Graphical programming of industrial applications • Software validation and verification of graphical code • Error handling of graphical code
4	Course Organization and Structure lecture (V) / laboratory (L)
5	Credits and Workload 2,5 CP / 75 hours in total, including 14 hours classroom teaching and lab. 1 SWS V / 1 SWS L
6	Examination Modalities Examination Prerequisites: None Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes.
7	Necessary Prerequisites None
8	Recommended Prerequisites Advanced Programming Techniques (MA04)
9	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03)
10	Applicability /Utilization All (software) design tasks; software and hardware architectures; mechatronics, WING. See appendix 2 BBPO (Compulsory options catalogues) for its suitability for other majors.
11	Literature • Dix, F.; Abowd, B.: Human-Computer Interaction, 3rd Edition, Pearson Education Ltd., 2004 978-0130461094 • Norman, D.: The Design of Everyday Things, revised and expanded edition, Basic Books, 2013, ISBN: 978-0-465-05065-9 • Scott MacKenzie: Human-Computer Interaction: An Empirical Research Perspective, 2013, Morgan Kaufmann; ISBN: 978-0124058651

MAwp06 Advanced Sensors for the Internet of Things

1	Module Name Advanced Sensors for the Internet of Things
1.1	Module Identifier MAwp06
1.2	Module Type Elective
1.3	Course Names Advanced Sensors for the Internet of Things - lecture
1.4	Semester 1 or 2
1.5	Module Responsible and Instructor Prof. Dr. Haid
1.6	Additional Instructors -
1.7	Study Program Master / Major Automation, Embedded
1.8	Teaching Language English
2	Module Content • The principles of sensor systems • Sensor limits and restrictions • Sensor validation • Sensor types and placement • Data analysis and filtering • Sensor algorithms • Condition monitoring • Localization and navigation systems • Deep Learning and Machine Learning • Blockchain and smart contracting
3	Learning Outcome / Competencies to know: • Develop sensor applications. to understand: • Design sensor systems. to apply:

	• Apply signal-processing methods and algorithms to transfer: • The practical aspects of sensor use and type • The environmental benefits of condition monitoring techniques
4	Course Organization and Structure lecture (V)
5	Credits and Workload 2,5 CP / 75 hours in total, including 28 hours classroom teaching and lab. 2 SWS V
6	Examination Modalities Examination Prerequisites: None Examination Type: Written exam covering the complete content of the module at the end of the semester. A make-up exam will be offered during the following semester. Examination Duration: 90 minutes.
7	Necessary Prerequisites None
8	Recommended Prerequisites None
9	Duration and Frequency of Course see module description of Elective 1 to 3 (MWP01, MWP02, MWP03)
10	Applicability /Utilization All (software) design tasks; software and hardware architectures; mechatronics, WING. See appendix 2 BBPO (Compulsory options catalogues) for its suitability for other majors.
11	Literature The following literature material will be provided: - Electronic lecture notes (excerpt of slides) - Workbook for the lab Further literature recommendations will be provided during the lecture.

Major Automation – general electives / electives from other majors

MGwp01	Research Project
MGwp02	Selected Research Topics
MGwp03	Ethics, Engineering, and Ecology
MCwp01	Digital Signal Processing Applications
MCwp02	Wireless Systems (Technologies)
MCwp03	Network Security
MCwp04	Mobile Communications
MCwp05	Optical Communications
MCwp06	(I)IoT and Cloud Networking
MCwp08	Image and Video Processing
MMwp02	Safety in Embedded Control Systems
MMwp04	Advanced Software Design Techniques
MMwp05	Security in Connected Embedded Systems
MMwp07	Advanced Driver Assistance Systems and Automated Driving
MPwp02	Automotive Electrical Power Train
MPwp03	Stationary & Mobile Energy Storage Systems
MPwp05	Switch Gear
MPwp06	Power Systems Planning
MPwp09	Hydrogen Technique and Fuel Cells